

SCIF2001 Research Roadmap

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October 2026

1 Literature Review

Quantum mechanics predicts correlations between measurements on entangled systems that cannot be reproduced by assigning independent, pre-existing properties to each particle. This issue was notably highlighted by the Einstein-Podolsky-Rosen argument, which questioned whether quantum mechanics provides a complete description of physical reality (Einstein et al. 1935). Bell later converted this into an experimentally testable problem by showing that any theory satisfying local realism must obey an upper bound on measurable correlations, whereas quantum mechanics predicts violations for suitable entangled states (Bell 1964).

The Clauser-Horne-Shimony-Holt (CHSH) formulation made Bell's theorem practical for laboratory experiments. In a polarization experiment, two measurement settings are selected on each photon path in an entangled photon-pair setup, a and a' for one photon, and b and b' for the other. Each polarisation measurement has two possible outcomes, conventionally labelled $+$ and $-$. For a pair of settings, the correlation is calculated from the coincidence counts as

$$E(a, b) = \frac{N_{++} + N_{--} - N_{+-} - N_{-+}}{N_{++} + N_{--} + N_{+-} + N_{-+}}. \quad (1)$$

The CHSH parameter is then

$$S = |E(a, b) - E(a, b') + E(a', b) + E(a', b')|. \quad (2)$$

Local hidden-variable theories require $S \leq 2$, while quantum mechanics predicts values as large as $2\sqrt{2}$ for an ideal maximally entangled state (Clauser et al. 1969). Therefore, a measured value significantly greater than 2 would provide evidence that the observed correlations cannot be explained by a local hidden-variable model. Correlation alone is not enough, since classically correlated sources can also produce similar outcomes. The relevant criteria is the violation of the CHSH bound.

Optical experiments provide a direct way to test this prediction. An entangled photon-pair source sends one photon into each measurement arm, with the measurement settings are defined by polarisers. A correlation device identifies detections occurring within a selected coincidence window, allowing the coincidence counts for each outcome to be measured. Early optical experiments established the feasibility of this approach. Freedman and Clauser measured linear-polarisation correlations from photons emitted in an atomic

cascade and found results consistent with quantum mechanics, but inconsistent with local hidden-variable restrictions (Freedman and Clauser 1972). Aspect, Dalibard, and Roger later measured photon-polarisation correlations using time-varying analysers and reported a violation of Bell’s inequality (Aspect et al. 1982). These experiments provide the experimental precedent for the proposed use of an entangled source, polarisers, and coincidence detectors.

A separate demonstration of the quantum nature of light is photon antibunching. The quantum theory of optical coherence describes multi-photon detection using correlation functions, including the second-order correlation function $g^{(2)}(\tau)$ (Glauber 1963). At zero delay, $g^{(2)}(0)$ measures the relative probability of detecting two photons close temporally. Ideal coherent light $g^{(2)}(0) = 1$, corresponding to Poissonian detection statistics, while thermal light generally exhibits bunching with $g^{(2)}(0) > 1$. Antibunched light satisfies $g^{(2)}(0) < 1$, indicating a suppression of near-simultaneous detections. The observation of antibunching in resonance fluorescence by Kimble, Dagenais, and Mandel was an important demonstration of nonclassical light because the measured photon statistics could not be explained by the relevant classical fluctuating-intensity models (Kimble et al. 1977).

Grangier, Roger, and Aspect demonstrated a related effect using a triggered photon source and a beam splitter. They observed strong anticorrelation between the two output detectors, consistent with a single-photon description and inconsistent with a classical wave model under the conditions of their experiment (Grangier et al. 1986). These results show that photon-timing measurements can provide practical evidence of nonclassical optical behaviour. However, there is an important distinction between actual photon antibunching measurements and a simple photon-pair correlation measurement. A reproducible coincidence peak from a heralded source demonstrates temporal correlation between the photons, but does not by itself establish $g^{(2)}(0) < 1$.

Modern loophole-free photon experiments require significant detector efficiency, independently selected measurement settings and additional control of locality and coincidence loopholes (Giustina et al. 2015). This means that due to the limitations of the equipment available, this experiment will be interpreted as a CHSH test under the equipment’s detection, coincidence selection, and fair sampling assumptions.

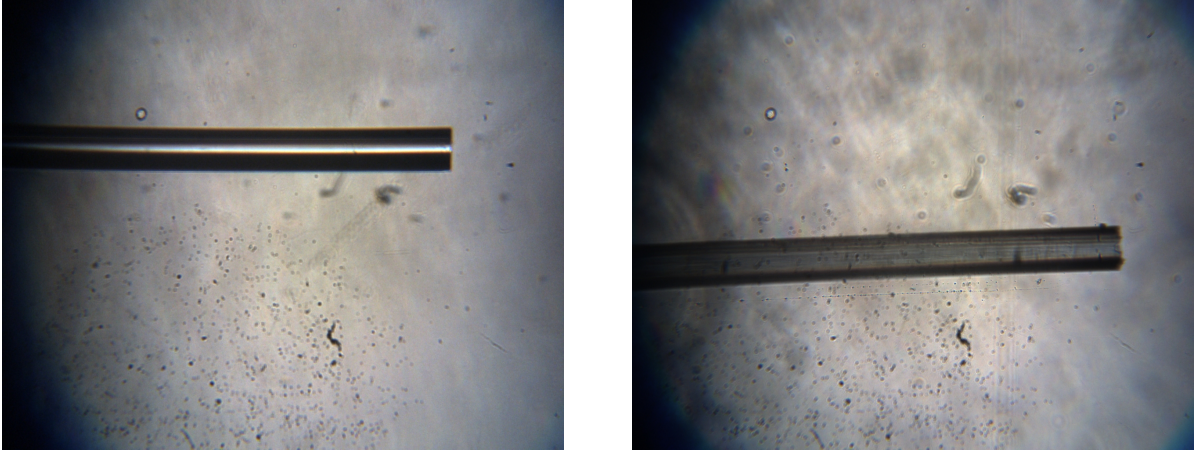
2 Skills Developed

2.1 Laboratory Skills

During the weeks leading up to this literature review, I developed the skills necessary to work with optical fibres in a laboratory environment. This included fibre preparation, laser alignment, and characterising both the quality of the alignment, as well as some properties of the fibre itself.

The fibre preparation process involved removing the protective coating and cladding, cleaning the exposed fibre with acetone, and cleaving the end of the fibre. Microscope inspection showed that defects at the cleaved surface can significantly reduce the amount of light coupled into the fibre (Fig. 1). This is relevant to the

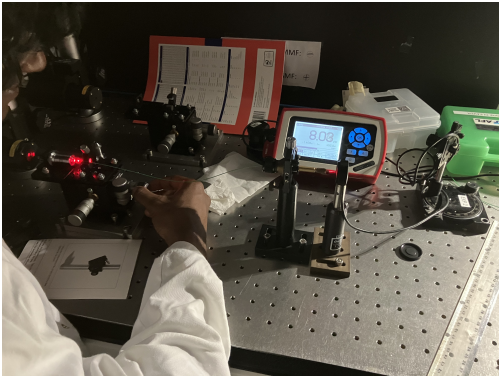
proposed photon experiment because coupling losses reduce the detected count rate and can make coincidence measurements more difficult.



(a) The laser-side of the final fibre we used to get 0.8 mW. (b) An example of our cleaves in the beginning. Notice the rough surface at the tip of the fibre, that is the actual fibre itself not any sort of dirt.

Figure 1: Two examples of cleaved optical fibre

After preparing the fibre, we aligned it to a laser using a microscope objective and three-axis position. The quality of the alignment was determined by maximising the optical power measured at the output of the fibre. An initial alignment produced a maximum power of $8.03\mu W$, while a later alignment produced $887.4\mu W$ (Fig. ??). This improvement demonstrated the sensitivity of fibre coupling to the cleave quality, fibre position, and objective focal length.



(a) The maximum amount of power achieved on the first day ($8.03\mu W$).



(b) The maximum amount of power achieved on the second day ($887.4\mu W$)

As a demonstration of a setup using optical fibre, we measured the angular acceptance of the fibre by sweeping the input angle and recording the transmitted power. A Gaussian fit (Fig. 3) to these measurements gave a numerical aperture of 0.364 ± 0.007 .

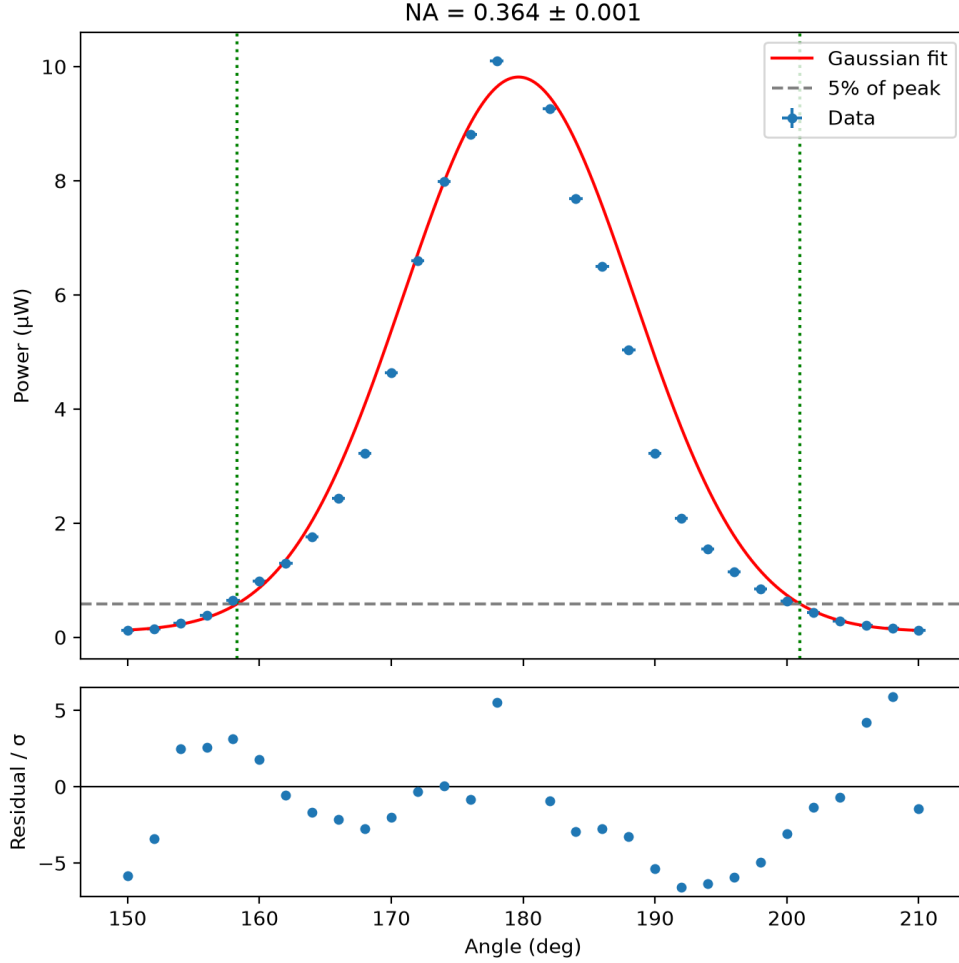


Figure 3: The Gaussian fit for the NA angle sweep. The likeliest cause of the poor fit is a combination of light leaking into the setup and incorrect placement of the tip of the fibre in the path of the laser.

These skills will be useful when aligning the photon-pair source, coupling light into fibres and maintaining stable optical paths. Efficient and stable coupling is particularly important because the proposed experiment relies on detecting a relatively small number of single-photon events and distinguishing actual coincidences from accidental detections.

2.2 Soft Skills

In addition to the skills learnt in the Higher-Year Lab, I was also able to develop more "soft skills" through the provided workshops. Specifically, I was able to improve my record-keeping, time management, and risk assessment and prevention.

The record-keeping skill in particular was useful, as it not only helped me keep a better record of the experiment I was doing, but it also meant that I was paying closer attention during the experiment as I

wanted to keep an accurate and detailed log of everything I did. Because of this, we were able to improve our methodology the second time we attempted to cleave the fibre and align the laser.

3 Proposed Methodology

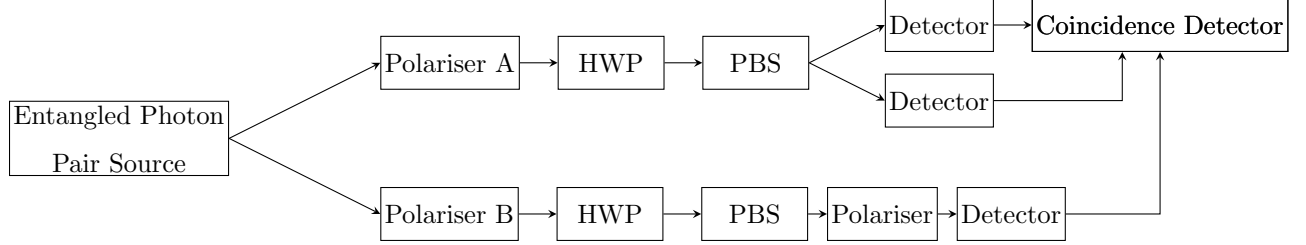


Figure 4: The proposed apparatus for the Bell experiment.

3.1 Proof of Concept

The first stage of the experiment will test the performance of the source, detectors, and correlation device. The entangled photon-pair source will be configured so that one detected photon provides a herald and the corresponding photon is sent to a second detector. The optical paths will initially be aligned using the available general optical equipment.

Before collecting the source data, detector dark counts and background counts will be measured with the source blocked. This will provide a baseline for estimating accidental coincidences. The source will then be operated for repeated acquisition intervals while the correlation device records the time difference between detection events.

The primary output will be histogram of detection delays. An actual heralded-pair signal should produce a coincidence peak above the surrounding accidental coincidence background. The peak position and width will be used to select an appropriate coincidence window. Repeated measurements will test whether the count rates and coincidence peak remains stable.

This provides three checks for the setup:

1. Whether the detectors produce stable count rate
2. Whether the correlation device resolve the relevant timing structure
3. Whether the chosen coincidence window is narrow enough to distinguish actual events from accidental coincidences.

If the optical arrangement allows for a $g^{(2)}(\tau)$ measurement, the normalised second-order correlation function will also be estimated. If the measured zero-delay value satisfies $g^{(2)}(0) < 1$, then we can say the photons are antibunched.

3.2 Polarisation-Correlation and CHSH Experiment

After the timing system has been validated, the two photons from the entangled source will be directed into separate measurement arms. Polarisers will be installed in both arms, with their angular positions recorded relative to a common reference, then split with a PBS.

Initial measurements will be performed at several analyser angles to verify the presence and orientation of the expected polarisation correlations. This step will also identify whether the source produces correlations described by $E(\alpha, \beta)$ or by the corresponding sign-reversed convention for the particular entangled state.

At each setting, the experiment will record detector count rate, coincidence counts, and the acquisition time. Background and accidental coincidences will be estimated either from off-peak timing windows or from measurements with one arm blocked, depending on the capabilities of the correlation device. The source alignment will be adjusted before collecting the final CHSH data.

The CHSH experiment will use two analyser settings on each measurement arm

$$(a, a'), \quad (b, b').$$

The four analyser-setting combinations

$$(a, b), \quad (a, b'), \quad (a', b), \quad (a', b')$$

will be measured. For each pair, the coincidence counts for the available $+$ and $-$ outcome channels will be collected over a fixed integration time. The correlation will then be calculated using Equation 1, which the four correlations will be substituted into Equation 2.

Because only three single-photon detectors are available, all four outcomes channels may not be recorded simultaneously. To combat this, a polariser will be placed in the B arm, and measurements will be inferred. This will introduce additional assumptions about detector stability and fair sampling, which is why this experiment is not claiming to be loophole-free.

Repeated acquisitions will be performed for each analyser setting. For approximately independent Poissonian counts, the uncertainty in an individual count will initially be estimated as

$$\sigma_N \approx \sqrt{N}.$$

These uncertainties will be propagated through the correlation and CHSH expressions to obtain

$$S = S_{\text{meas}} + \sigma_S.$$

A violation will be considered meaningful only if $S_{\text{meas}} > 2$ by a statistically significant margin and the result is reasonably consistent across repeated acquisitions.

3.3 Limitations and Contingency

The main experimental risks are low photon-pair rates, detector inefficiency, accidental coincidences, unstable optical alignment, and incomplete access to the four required outcome channels.

If the coincidence peak is weak, the alignment will be checked and the integration time increased. If accidental coincidences are significant, the coincidence window will be reduced where this does not exclude actual photon pairs. If the result varies between repeated measurements, the source alignment, fibre coupling, and detector count rates will be monitored before additional data are collected.

If the apparatus cannot provide all outcome channels required for a complete CHSH analysis, the project will report the available polarisation-correlation visibility and compare its angular dependence with the quantum prediction. If the CHSH parameter cannot be estimated reliably, the heralded-correlation proof of concept will remain the primary experimental outcome.

4 Project Plan

In order to maintain a schedule that aligns with the outcomes and due dates of the course, I am attempting to adhere to the following plan to meet these outcomes and dates. This timeline takes into account personal events, study load for other courses, and a small amount of built-in leniency in the (likely) event of not meeting these targets.

Week 5: Read over background material for both the milestone and main experiment. Begin to prepare a plan for once we have access to the lab.

Week 6: Finalise the plan for the experiments and finish up any reading left to do. Read over the equipment manuals so I know I understand how the equipment works before using it.

Week 7: Prepare the experimental setup, and if possible perform the proof-of-concept experiment so we can ensure everything is working correctly.

Week 8: Perform the actual experiment and record all the data.

Week 9: Verify results and draw conclusions, good or bad, regarding the outcome of the experiment. If required, perform the experiment again to get more data.

Week 10: Prepare the final presentation with everything we have done.

Week 11: Prepare the video.

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